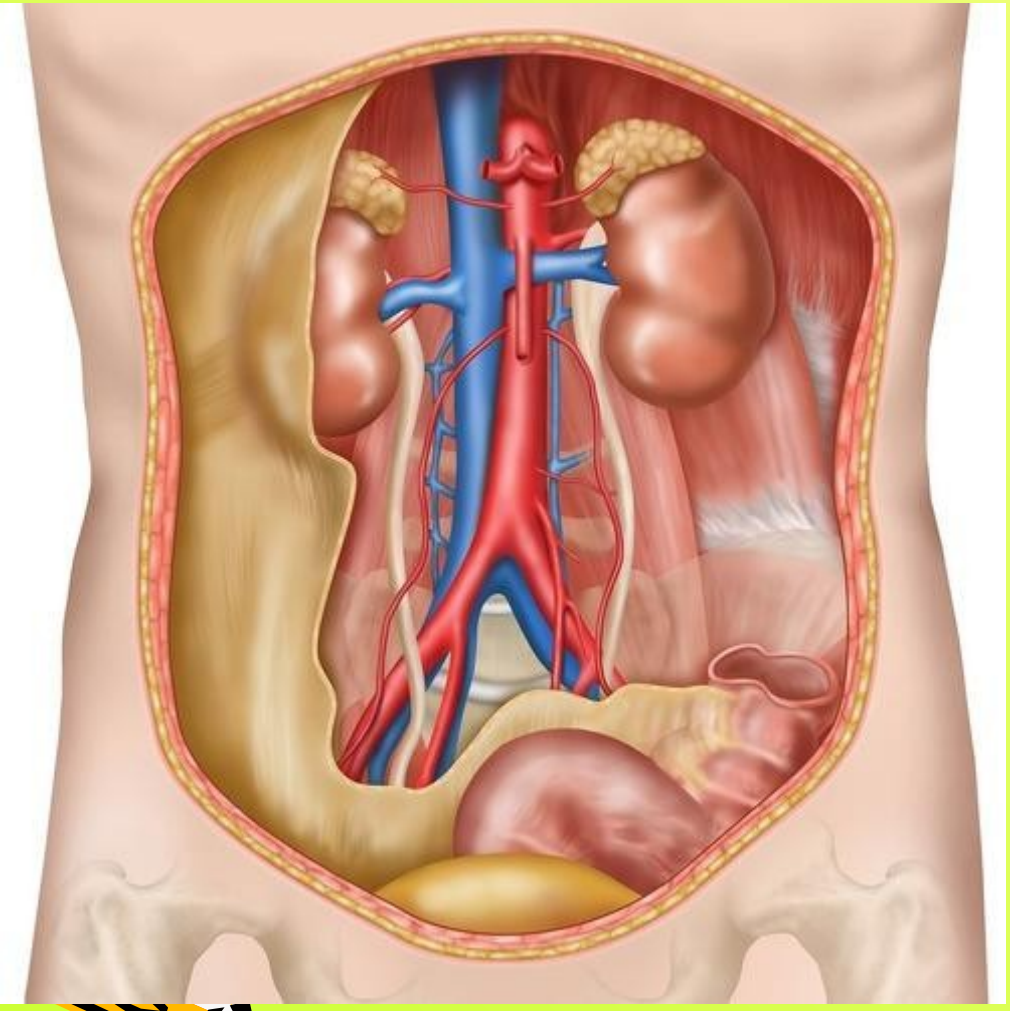
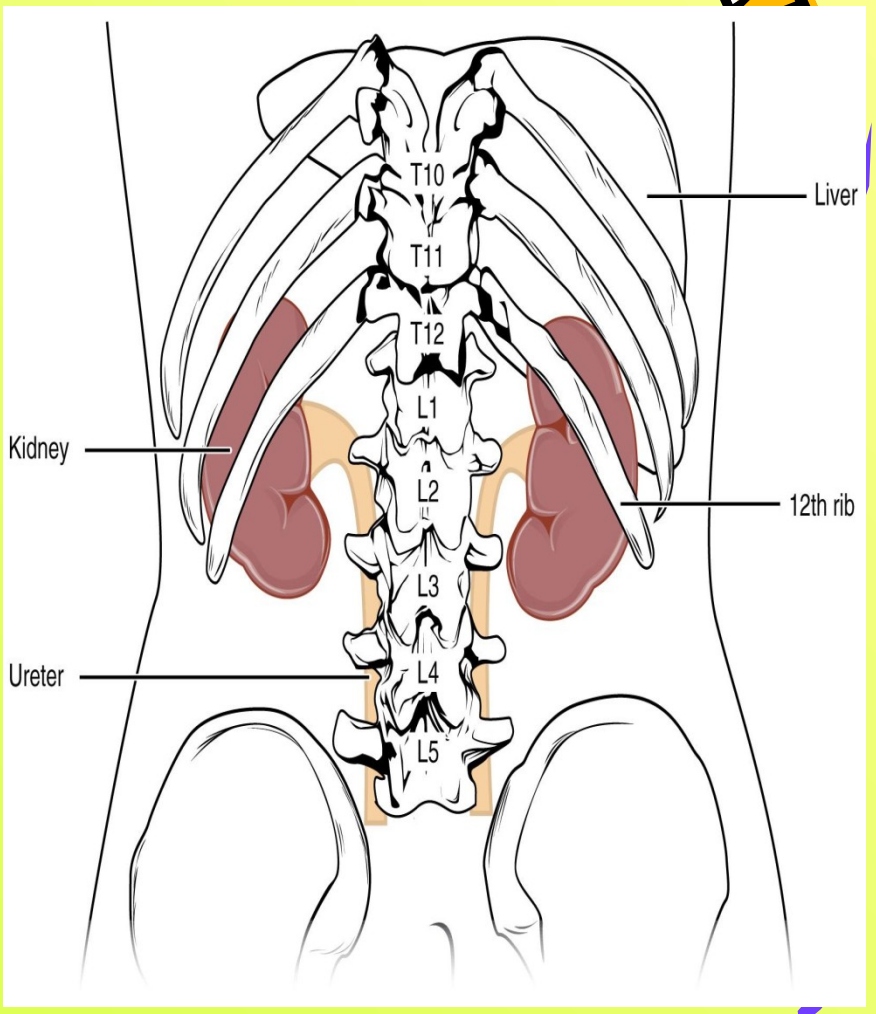
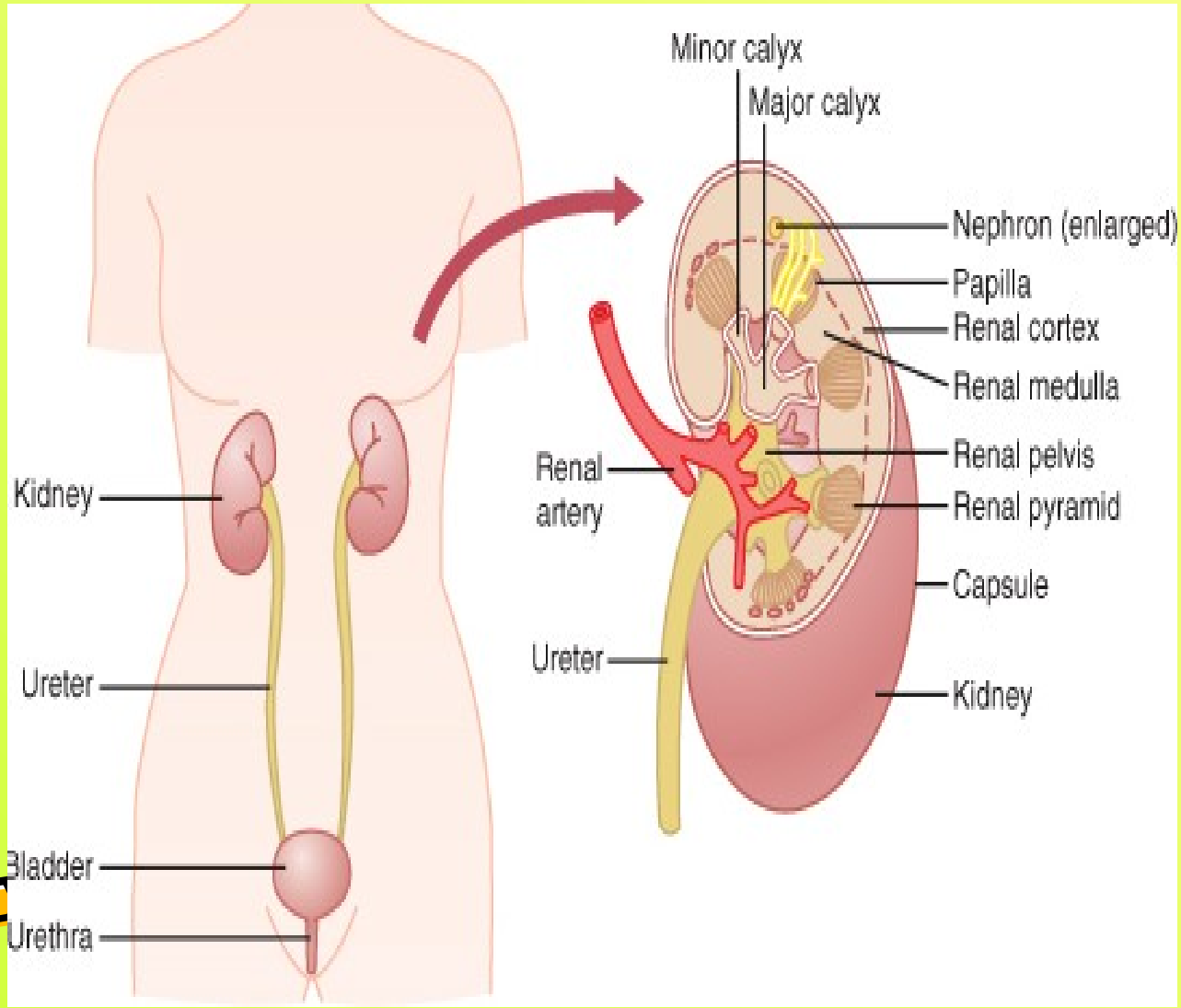
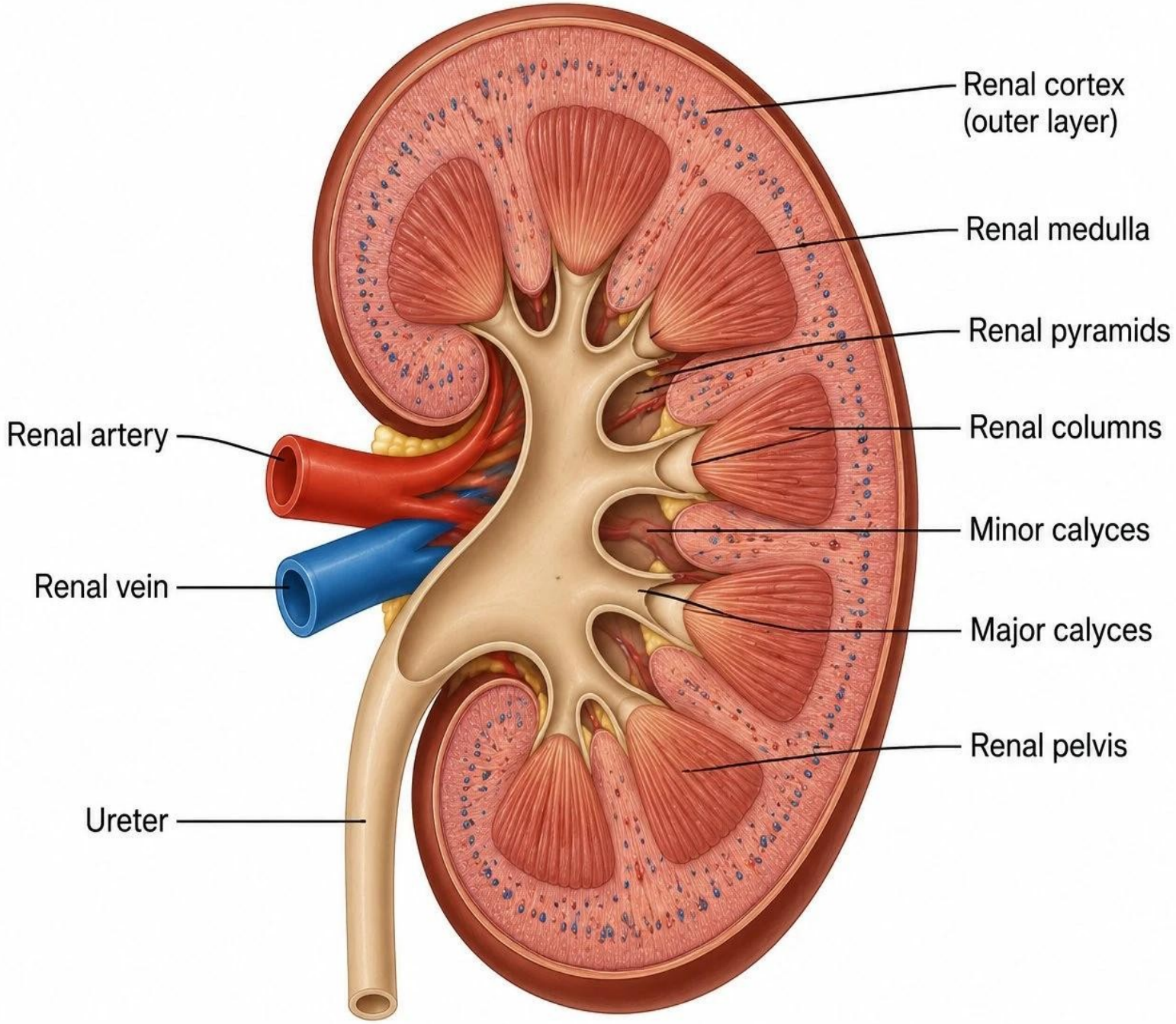
Renal Physiology: Urinary System





Gross Anatomy




Renal cortex
(outer layer)
Renal medulla
Renal pyramids
Renal artery
Renal columns
Minor calyces
Renal vein
Major calyces
Renal pelvis
Ureter

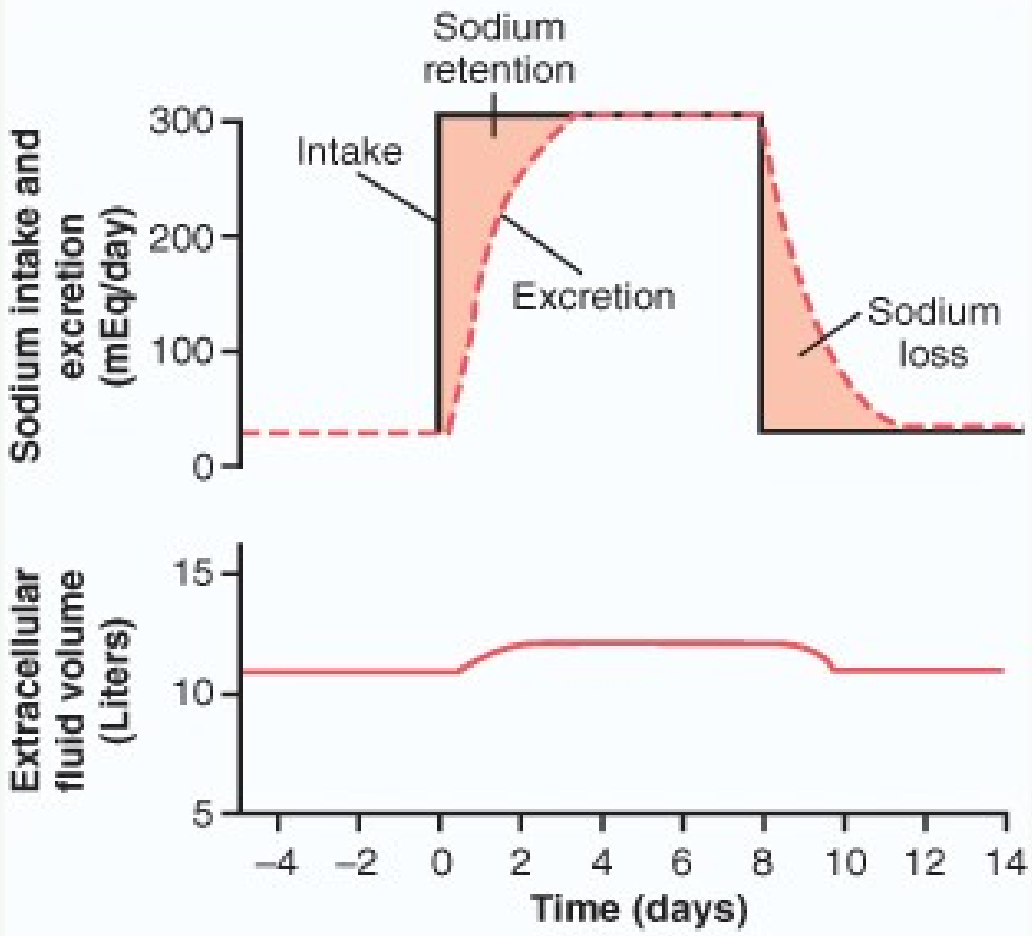
Physiological Roles

- Regulation of body fluid osmolality & volumes
- Regulation of electrolyte balance
- Regulation of acid – base balance
- Excretion of metabolic products & foreign substances
- Production and secretion of hormones



TABLE 38–1 Typical urinary and plasma concentrations of some physiologically important substances.

Substance	Concentration in		U/P Ratio
	Urine (U)	Plasma (P)	
Glucose (mg/dL)	0	100	0
Na^+ (mEq/L)	90	140	0.6
Urea (mg/dL)	900	15	60
Creatinine (mg/dL)	150	1	150


Sodium retention
Intake
Excretion
Sodium loss
Sodium intake and excretion (mEq/day)
300
200
100
0
Extracellular fluid volume (Liters)
15
10
5
–4
–2
0
2
4
6
8
10
12
14
Time (days)

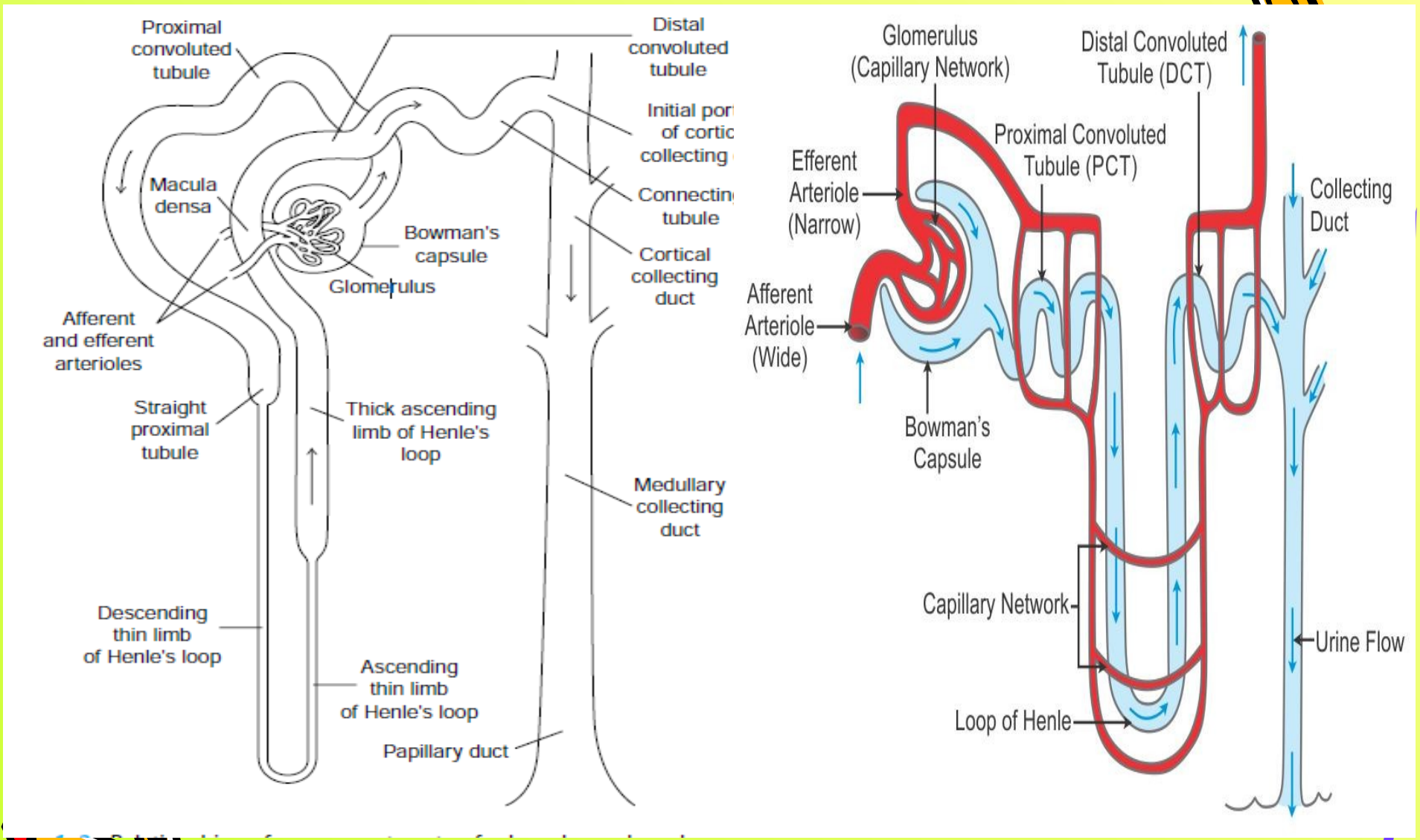
Ultrastructure of the Nephron

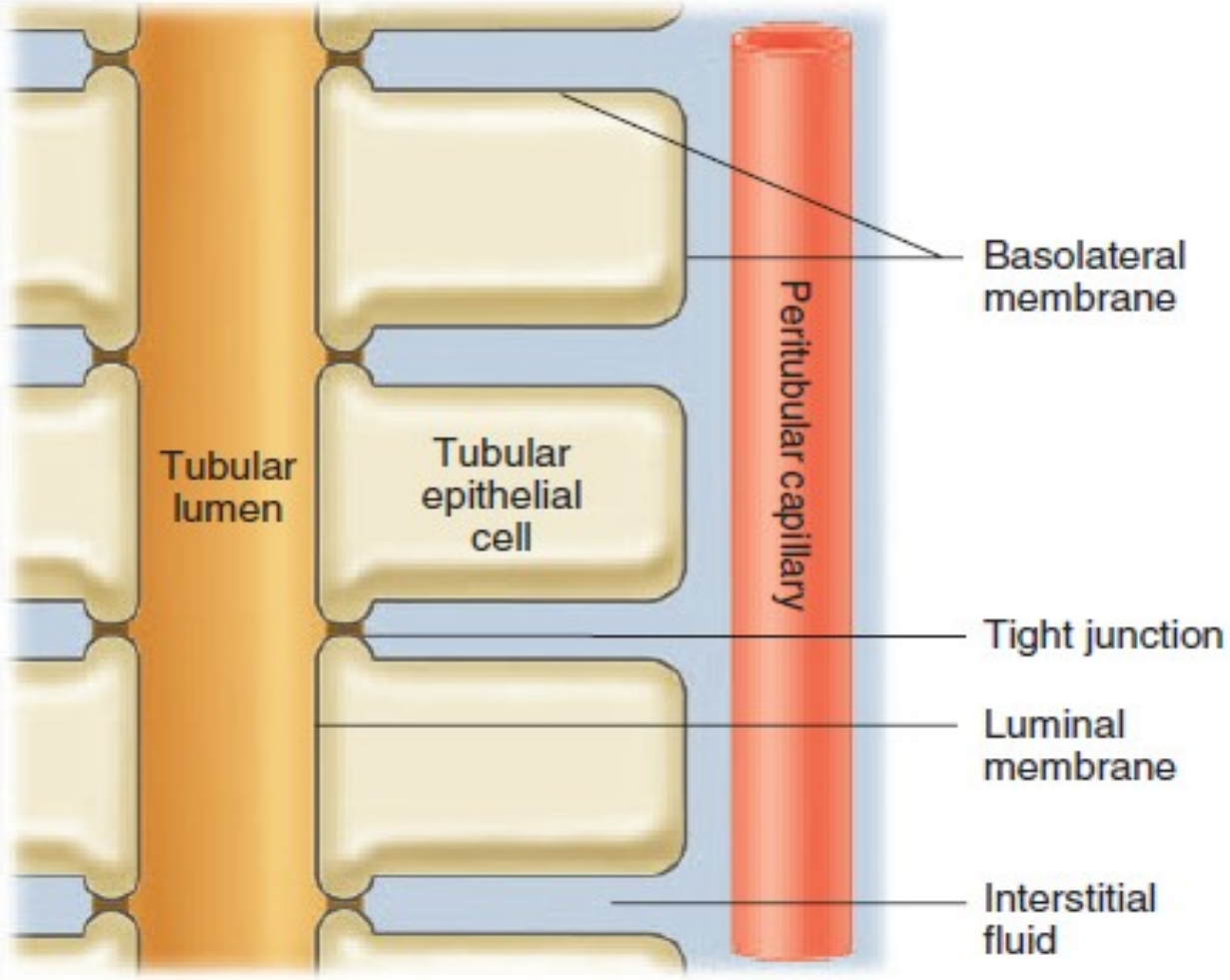


- Renal Corpuscle
- Proximal Tubule
- Loop of Henle
- Distal Tubule
- Collecting Duct System: principal cells (P cells) and intercalated cells (I cells)
- 3 types of Nephrons: Superficial & Juxtamedullary



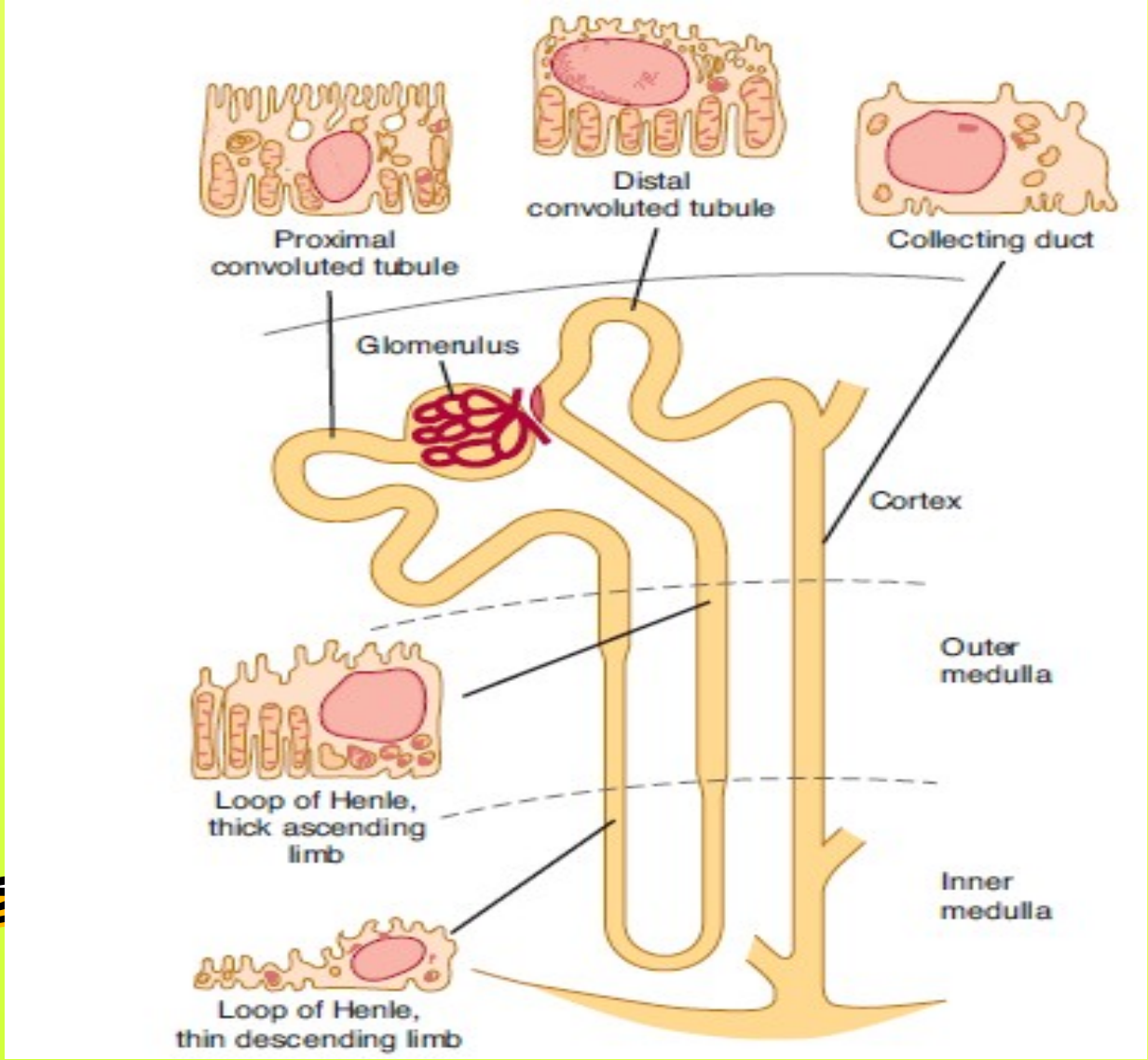
Table 1-1. Terminology for the Tubular Segments	
Sequence of segments	**Combination terms used in text**
Proximal convoluted tubule	Proximal tubule
Proximal straight tubule	
Descending thin limb of Henle's loop	Henle's loop
Ascending thin limb of Henle's loop	
Thick ascending limb of Henle's loop (contains macula densa near end)	
Distal convoluted tubule	
Connecting tubule	Collecting-duct system
Cortical collecting duct	
Outer medullary collecting duct	
Inner medullary collecting duct (last portion is papillary duct)	

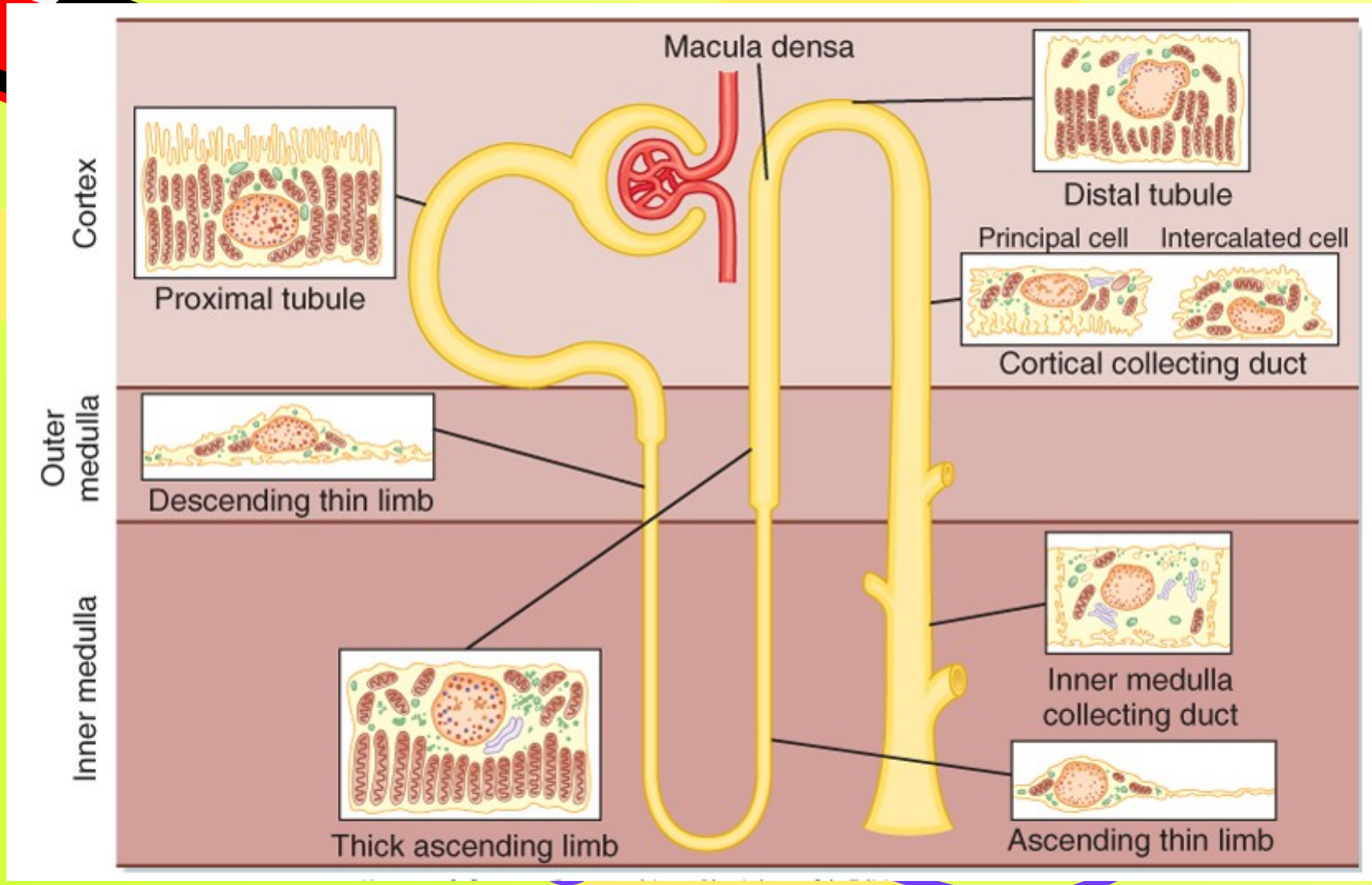

Proximal convoluted tubule
Distal convoluted tubule
Initial por
of cortic
collecting
Macula densa
Connectin
tubule
Bowman's capsule
Cortical collecting duct
Glomerulus
Afferent and efferent arterioles
Straight proximal tubule
Thick ascending limb of Henle's loop
Medullary collecting duct
Descending thin limb of Henle's loop
Ascending thin limb of Henle's loop
Papillary duct
Glomerulus (Capillary Network)
Distal Convoluted Tubule (DCT)
Proximal Convoluted Tubule (PCT)
Efferent Arteriole (Narrow)
Collecting Duct
Afferent Arteriole (Wide)
Bowman's Capsule
Capillary Network
Urine Flow
Loop of Henle



Diagrammatic representation of tubular epithelium.

Diagram of a juxtamedullary nephron

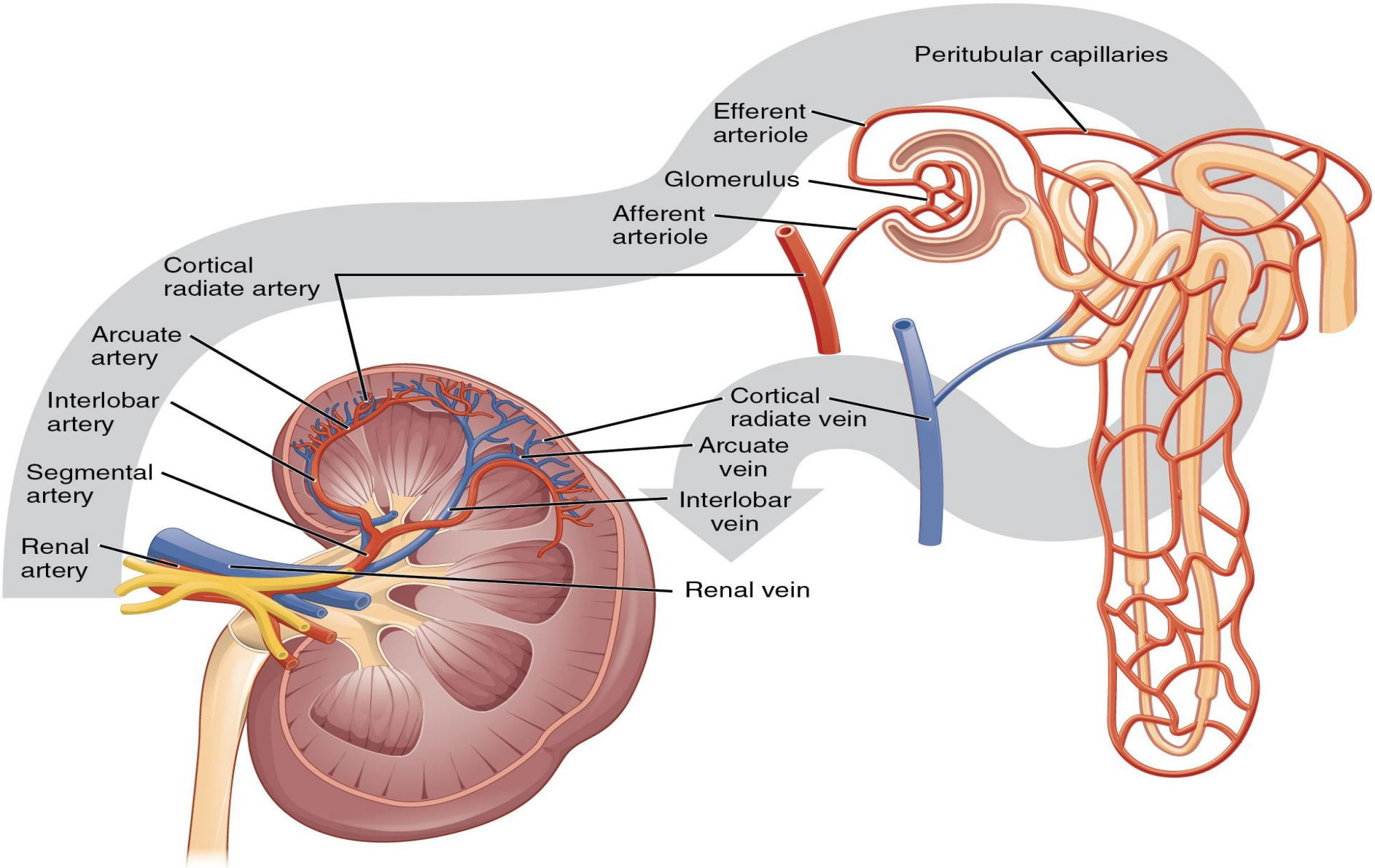


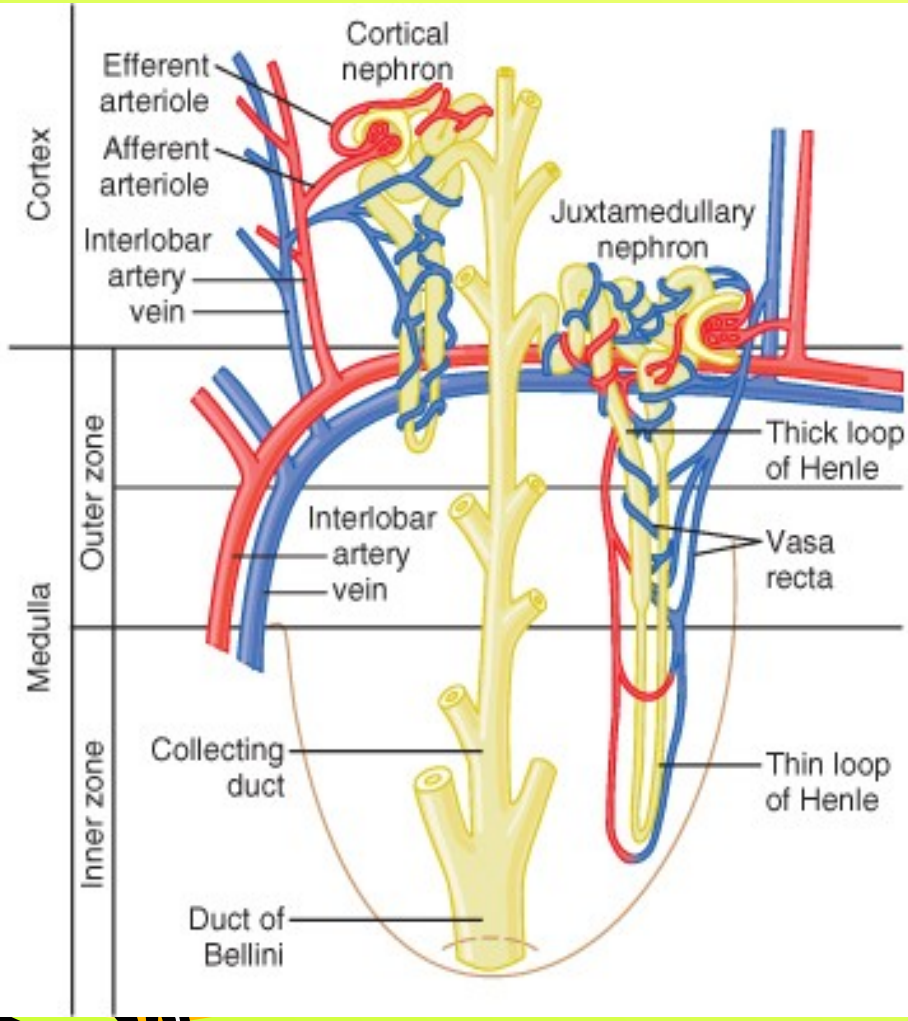


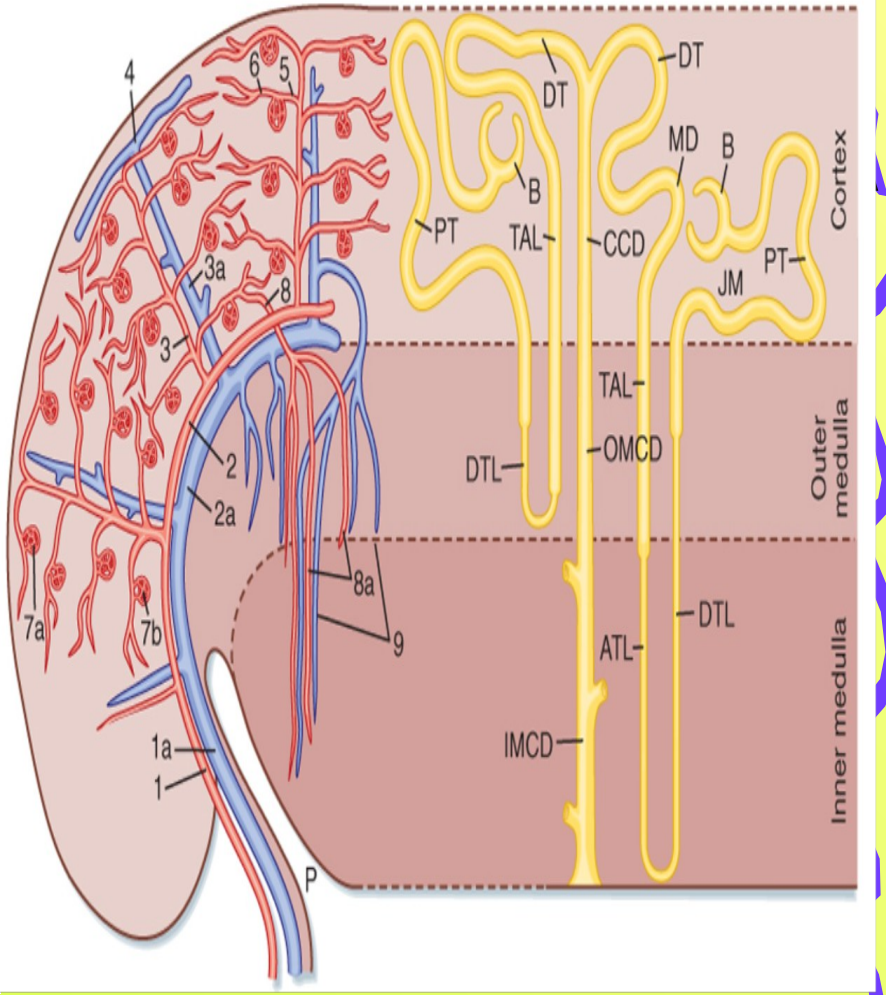
Tamm Horsfall Protein

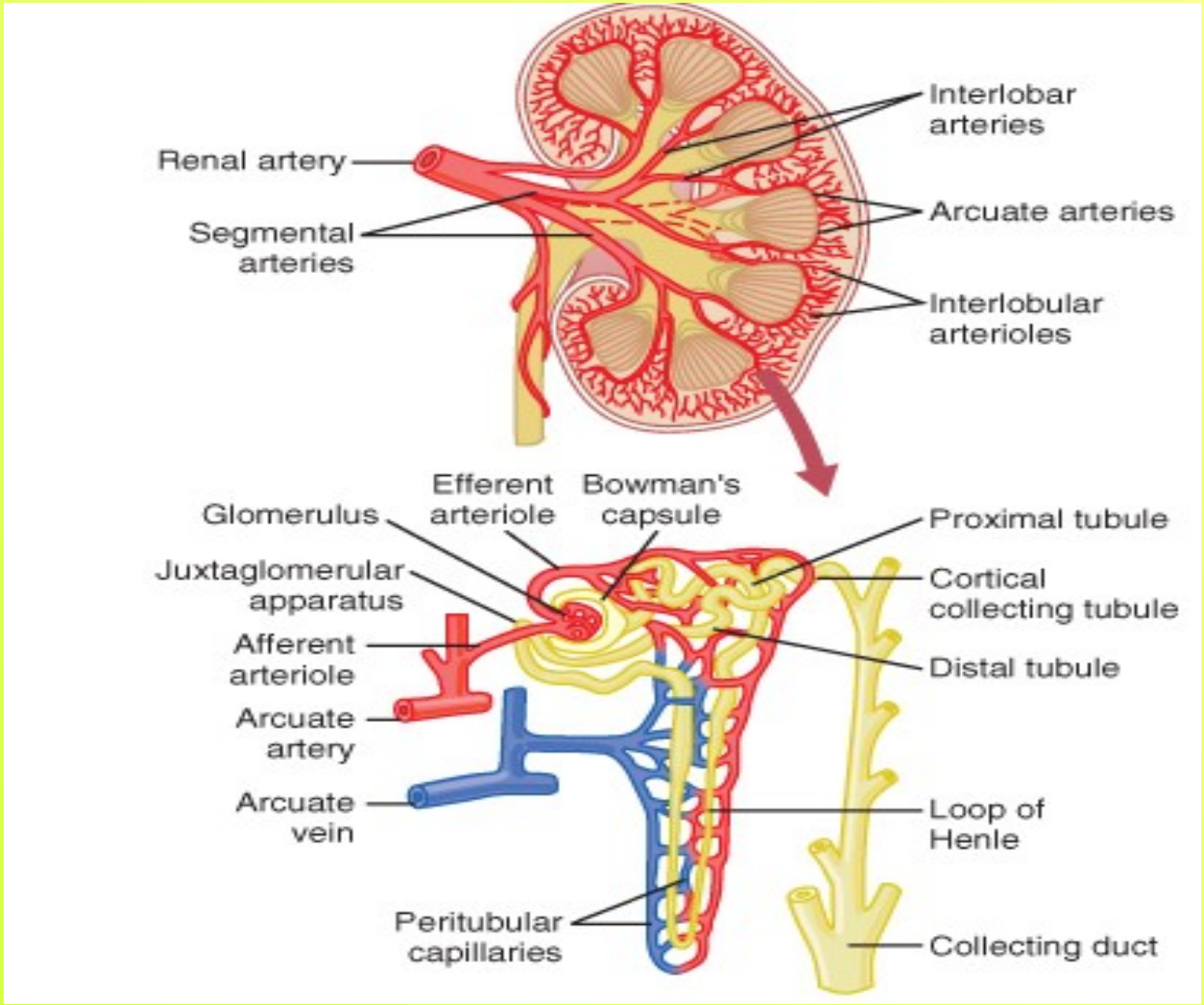
Urodilatin

Renal Vasculature




Cortex
Medulla
Outer zone
Inner zone
Efferent arteriole
Afferent arteriole
Interlobar artery
vein
Cortical nephron
Juxtamedullary nephron
Thick loop of Henle
Vasa recta
Interlobar artery
vein
Collecting duct
Thin loop of Henle
Duct of Bellini


4
6
5
3a
8
3
2
2a
7a
7b
8a
9
1a
1
P
DT
B
PT
TAL
CCD
MD
JM
DTL
OMCD
ATL
IMCD
Cortex
Outer medulla
Inner medulla


Renal artery
Segmental arteries
Interlobar arteries
Arcuate arteries
Interlobular arterioles
Glomerulus
Efferent arteriole
Bowman's capsule
Juxtaglomerular apparatus
Afferent arteriole
Arcuate artery
Arcuate vein
Peritubular capillaries
Proximal tubule
Cortical collecting tubule
Distal tubule
Loop of Henle
Collecting duct

Renal Blood Flow(RBF)

- Arteries
- afferent arterioles
- efferent arteriole
- peritubular capillaries
- **vasa recta:**

1- conveying oxygen and important nutrients to nephron segments.

2-delivering substances to the nephron for secretion.

3-serving as a pathway for the return of reabsorbed water and solutes to the circulatory system

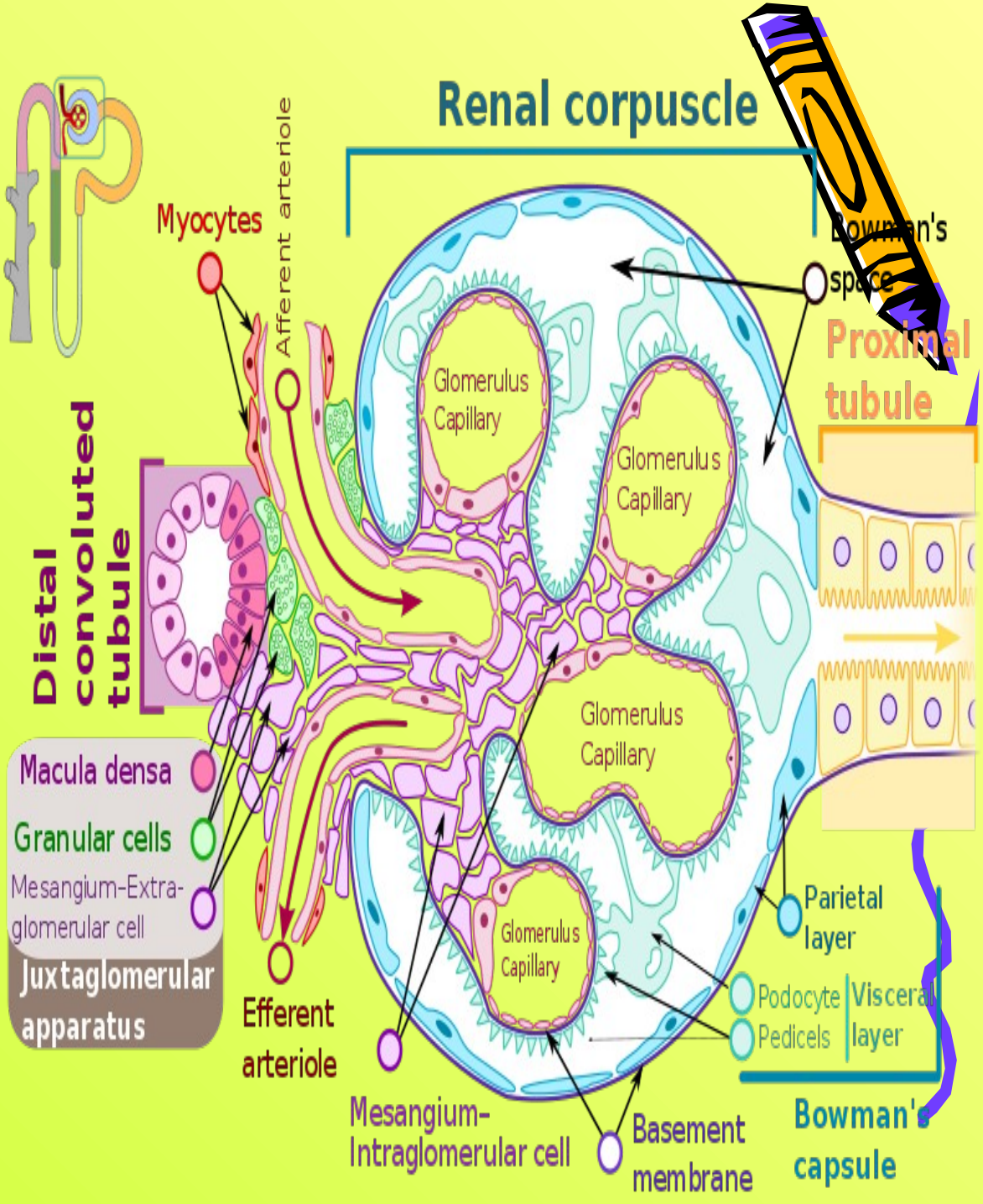
4- concentrating and diluting the urine.

-Mesangium:

1) structural support
2) secrete the extracellular matrix
3) phagocytic activity
4) secrete prostaglandins and proinflammatory cytokines
5) influence the GFR

-Extraglomerular mesangial cells





Juxtaglomerular Apparatus

- Macula Densa
- Extraglomerular mesangial cells(Lacis)
- Granular cells:

modified smooth muscle cells that manufacture, store, and release renin.

]GA & tubuloglomerular feedba mechanism



Fig 4. **Juxtaglomerular apparatus**

